

**MATH 1342 – Elementary Statistics – Week 8: Percentile Intervals and Introducing Hypothesis Tests**

**Oct 13 – 15** | 3.4 Bootstrap Confidence Intervals Using Percentiles • 4.1 Introducing Hypothesis Tests | Fall 2026 • Section 036 • TTh 6:00–7:20 PM (DIL)

| Course-level Objectives                                                                                                                                                                                                                                                                                                                                                                                                                                   | Module-level Objectives                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Learning Materials                                                        | #             | Activities                                                                                                                            | #             | Assessments                                                                                      | #             |
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| 3. Find and use empirical probabilities in bootstrap distributions to find confidence intervals and in randomization distributions to test hypotheses.<br>9. Estimate single means, difference of two means, single proportions and difference of two proportions using confidence intervals. Interpret the results.<br>10. Demonstrate skills in hypothesis testing for means and proportions, for single populations and comparison of two populations. | 1. Determine which percentiles of a bootstrap distribution bound a 90%, 95%, or 99% confidence interval, and construct the interval with StatKey. (CO3, CO9)<br>2. Predict how changing the confidence level or the sample size affects the width of an interval. (CO9)<br>3. Judge whether a bootstrap confidence interval is trustworthy from the shape of the bootstrap distribution and whether the sample is representative. (CO3)<br>4. Identify the parameter of interest and write null and alternative hypotheses from a research question. (CO10)<br>5. Classify a test as right-tailed, left-tailed, or two-tailed, and explain the logic of assuming $H_0$ and asking whether the data are surprising. (CO10) | e-Text: 3.4 Bootstrap Confidence Intervals Using Percentiles              | MO1, MO2, MO3 | Participation: Percentile Confidence Intervals – StatKey: 90/95/99% intervals for the same data; compare SE and percentile methods    | MO1, MO2, MO3 | Online Homework: 3.4 Bootstrap Confidence Intervals Using Percentiles (due Sun Oct 18, 11:59 PM) | MO1, MO2, MO3 |
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